



# University of Engineering & Technology Peshawar

Programming Fundamentals (C++) Lab

## LAB REPORT 2

|                     |                                                          |
|---------------------|----------------------------------------------------------|
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| <b>COUSRE:</b>      | <b>EE-170L COMPUTER<br/>PROGRAMMING LAB</b>              |
| <b>DATE:</b>        | <b>3/6/26</b>                                            |
| <b>LAB TITLE:</b>   | <b>C++ DATA TYPE, MEMORY SIZE<br/>AND MEMORY ADDRESS</b> |

**SUBMITTED TO ENGR RIZWAN ULLAH**

### Learning Objectives:

The objective of this lab is:

- To understand different data types in C++
- To learn how to declare variables



- To observe the memory size of different data types
- To find the memory addresses of variables using the address operator (&)

### **Theory:**

In C++, variables are used to store data in memory. Each variable has:

- Data Type – defines what kind of data it can store.
- Memory Size – the amount of memory allocated for the variable.
- Memory Address – the location where the variable is stored in RAM.

| Data Type | Description                  |
|-----------|------------------------------|
| int       | Stores integer numbers       |
| float     | Stores decimal numbers       |
| double    | Stores large decimal numbers |
| char      | Stores a single character    |
| bool      | Stores true or false         |



### **Algorithm:**

1. Start the program
2. Declare variables of different data types
3. Assign values to variables
4. Print the values of variables
5. Use sizeof() to print memory size
6. Use & operator to print memory addresses
7. End the program

### **Program code:**

```
#include <iostream>
using namespace std;
int main() {
    int a = 10;
    float b = 5.5;
    double c = 20.99;
    char d = 'A';
    bool e = true;

    cout << "VALUES:\n";
    cout << "int a = " << a << endl;
```



```
cout << "float b = " << b << endl;
cout << "double c = " << c << endl;
cout << "char d = " << d << endl;
cout << "bool e = " << e << endl;

cout << "\nMEMORY SIZE (in bytes):\n";
cout << "Size of int: " << sizeof(a) << endl;
cout << "Size of float: " << sizeof(b) << endl;
cout << "Size of double: " << sizeof(c) << endl;
cout << "Size of char: " << sizeof(d) << endl;
cout << "Size of bool: " << sizeof(e) << endl;

cout << "\nMEMORY ADDRESS (Location):\n";
cout << "Address of a: " << &a << endl;
cout << "Address of b: " << &b << endl;
cout << "Address of c: " << &c << endl;
cout << "Address of d: " << (void*)&d << endl;
cout << "Address of e: " << &e << endl;

return 0;
}
```



## **Program Output**

VALUES:

int a = 10

float b = 5.5

double c = 20.99

char d = A

bool e = 1

MEMORY SIZE (in bytes):

Size of int: 4

Size of float: 4

Size of double: 8

Size of char: 1

Size of bool: 1

MEMORY ADDRESS (Location):

Address of a: 0x61ff08

Address of b: 0x61ff04

Address of c: 0x61fef8

Address of d: 0x61fef7

Address of e: 0x61fef6



## **RESULT:**

The program successfully displayed;

- The program successfully displayed:
- Values of different variables
- Memory size of each data type
- Memory addresses of variables

## **Conclusion:**

From this experiment we learned;

- Different **C++ data types store different kinds of data**
- Each data type occupies **different memory sizes**
- The **address operator (&)** can be used to find the memory location of variables